Part II

Finite Volume Methods

Chapter 8

Nonlinear Hyperbolic Equations

If we set $f(u) = au$, the linear advection equation can be written as

$$u_t + f(u)_x = 0$$

This is the general form of a conservation law. In particular, if $f(u)$ is some nonlinear function of u we have a nonlinear conservation law. Finite Volume Methods are particular popular for such nonlinear equations. Therefore, we make the transition from finite difference methods to finite volume methods at the same time we make the transition from linear to nonlinear equations. As we shall see that transition is made easier by the fact that in the setting we have used in chapter 6 (i.e 1D Cauchy problems) the difference between finite volume methods and finite difference methods is largely notional. Before we begin, however, we motivate conservation laws by considering a particular example.

8.1 Motivation: The Euler Equations

We consider a compressible fluid. Conservation laws for mass, momentum, and energy may be used to derive the field equations governing fluid flow. Here, we do not consider boundary conditions, and simply assume that the entire three-dimensional space $\mathbb{R}^3$ is filled with a compressible fluid. We derive an expression of the rate-of-change of the above-mentioned conserved quantities in an arbitrary control volume $\Omega \subset \mathbb{R}^3$. Fig. 8.1 shows an illustration. For simplicity, we assume the control volume is stationary with respect to some reference frame. The differential equations obtained at the end of this section are independent of whether or not the control volume is stationary.

Let the local fluid velocity vector be given by $\mathbf{v}(\mathbf{x}) = (v_1(\mathbf{x}), v_2(\mathbf{x}), v_3(\mathbf{x}))^T$, and denote the density (or specific mass) by $\rho(\mathbf{x})$. Here $\mathbf{x} = (x_1, x_2, x_3)^T$ is the coordinate vector with respect to our reference frame[1]. Conservation of mass is expressed by

$$\frac{d}{dt}\int_\Omega \rho \, dV = -\int_{\partial\Omega} \rho \mathbf{v} \cdot \mathbf{n} \, dA \ , \tag{8.1}$$

which states that the rate of change of mass inside a control volume is due solely to mass flux across its boundary. Figure 8.2 shows a graphical illustration of the derivation of the right-hand side.

According to Newton's second law of motion, the rate of change of momentum is equated with the external forces acting on the fluid. Such forces are given by thermodynamic pressure. We model the fluid as *inviscid*, and so neglect shear forces. For simplicity, we also neglect body forces, such as gravity, and obtain for the specific momentum ρv_i,

$$\frac{d}{dt}\int_\Omega \rho v_i \, dV = -\int_{\partial\Omega} \rho v_i \mathbf{v} \cdot \mathbf{n} \, dA - \int_{\partial\Omega} p n_i \, dA \ , \qquad i = 1, 2, 3 \ , \tag{8.2}$$

[1] In the following we often suppress in our notation the dependence on $\mathbf{x}$ of the quantities under investigation.

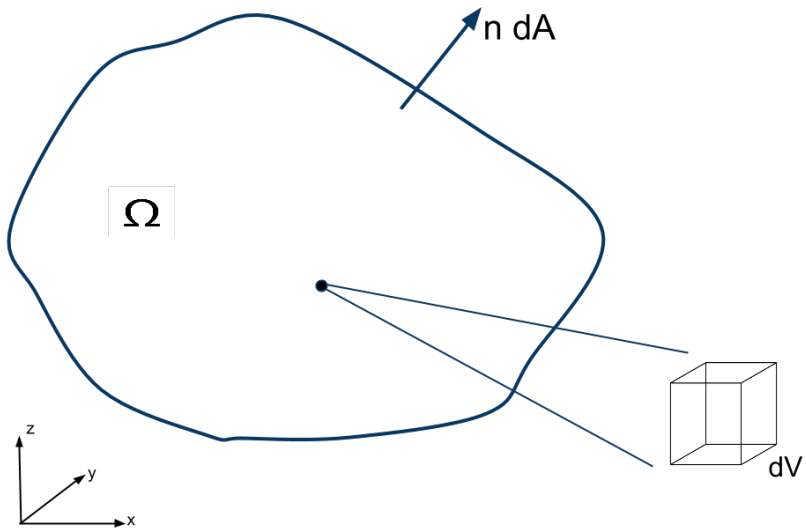

Figure 8.1: Control volume and infinitesimal fluid element.

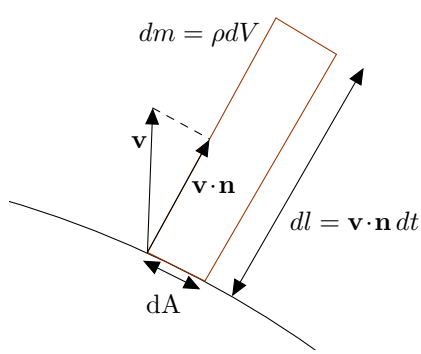

Figure 8.2: The Fluid leaving the control volume during the time interval dt sweeps out a small volume element $dV = dAdl$, where $dl = \mathbf{v}\cdot\mathbf{n}dt$. The mass contained in that volume is given by $dm = \rho\mathbf{v}\cdot\mathbf{n}dAdt$. Dividing by dt and integrating over the surface area gives the *mass flux* across the boundary of the control volume.

where the first term on the right-hand side again models loss or gain of momentum by fluid entering or leaving the domain. The second term models the forces exerted on the control volume by the pressure p (which by definition acts *onto* the fluid, i.e. in direction opposite to the outward pointing normal $\mathbf{n}$).

The total energy per unit volume is written as $E = \rho e_{int} + \frac{1}{2}\rho \mathbf{v} \cdot \mathbf{v}$, where the second term is mechanical energy, while e_{int} stands for internal energy. The amount of energy inside the control volume may change for a variety of reasons, including transport across the boundaries, and work done by the forces acting on the boundary. We could also consider heat conduction, but since we have neglected viscous effects before, we shall not include heat conduction[2]. We also neglect other contributions, such as radiation or heat sources. The energy balance is thus given by

$$\frac{d}{dt}\int_\Omega E\, dV = -\int_{\partial\Omega} (E+p)\mathbf{v}\cdot\mathbf{n}\, dA. \tag{8.3}$$

We are thus in a position to write the conservation laws for mass, momentum, and energy for an inviscid compressible fluid in *integral form*:

$$\frac{d}{dt}\int_\Omega \rho\, dV + \int_{\partial\Omega} \rho\mathbf{v}\cdot\mathbf{n}\, dA = 0 \tag{8.4}$$

$$\frac{d}{dt}\int_\Omega \rho\mathbf{v}\, dV + \int_{\partial\Omega} (\rho\mathbf{v}\otimes\mathbf{v} + pI)\mathbf{n}\, dA = 0 \tag{8.5}$$

$$\frac{d}{dt}\int_\Omega E\, dV + \int_{\partial\Omega} (E+p)\mathbf{v}\cdot\mathbf{n}\, dA = 0\ , \tag{8.6}$$

where $\mathbf{v}\otimes\mathbf{v} = (v_i v_j)_{i,j=1,2,3}$ is the tensor formed by the dyadic product of $\mathbf{v}$ with itself, and I is the unit tensor. These are the *Euler* equations in integral form. The equations in differential form are obtained by applying the divergence theorem, thus converting all integrals to volume integrals over Ω. However, noting that the control volume Ω is arbitrary, we may drop the volume integrals, and require all equalities to hold for the integrands. This leads to

$$\frac{\partial\rho}{\partial t} + \nabla\cdot(\rho\mathbf{v}) = 0\ , \tag{8.7}$$

$$\frac{\partial\rho\mathbf{v}}{\partial t} + \nabla\cdot(\rho\mathbf{v}\otimes\mathbf{v} + pI) = 0\ , \tag{8.8}$$

$$\frac{\partial E}{\partial t} + \nabla\cdot(\mathbf{v}\,(E+p)) = 0\ . \tag{8.9}$$

Note that these equations are not closed yet, as we have six unknowns, but only five equations. We come back to this issue shortly. First, however, introduce the general notation for a *conservation law*:

$$\frac{\partial\mathbf{u}(x,t)}{\partial t} + \sum_{i=1}^{d}\frac{\partial\mathbf{f}_i(\mathbf{u})}{\partial x_i} = 0, \tag{8.10}$$

where f is called the *flux vector*. Equations that can be written in the form of eq. (8.10) are called *conservation laws*.

Example 8.1 For the Euler equations (8.7)-(8.9) the state and flux vector are given by

$$\mathbf{u} = \begin{pmatrix} \rho \\ \rho v_1 \\ \rho v_2 \\ \rho v_3 \\ E \end{pmatrix}, \qquad f_i = \begin{pmatrix} \rho v_i \\ \rho v_1 v_i + p\delta_{1i} \\ \rho v_2 v_i + p\delta_{2i} \\ \rho v_3 v_i + p\delta_{3i} \\ (E+p)v_i \end{pmatrix}, \qquad i = 1,2,3\ , \tag{8.11}$$

[2]Heat conduction is a viscous effect: At the molecular level, heat conduction is diffusion of kinetic energy associated with the thermal, random motion of the molecules.

To close the system (8.7)-(8.9) one needs an equation of state, which relates the pressure and temperature to the other variables. For many compressible fluids under moderate conditions, the *ideal gas law* holds. For a *termally* perfect gas one may write

$$p = \rho R T, \tag{8.12}$$

where $R = c_p - c_v$ is the specific gas constant, and c_p and c_v are the specific heats at constant pressure and volume, respectively. For a *calorically* perfect gas the specific heats are constant, and there holds

$$e_{int} = c_v T.$$

Combining these relations leads to

$$\begin{aligned} p &= \rho R T, \\ p &= \rho (c_p - c_v) \frac{e_{int}}{c_v}, \\ p &= (\gamma - 1) \left\{ E - \frac{1}{2} \rho \mathbf{u} \cdot \mathbf{u} \right\} , \end{aligned} \tag{8.13}$$

where the definition of the total energy has been used. Furthermore, we have introduced the ratio of specific heats $\gamma = c_p / c_v$. (Note that $\gamma > 1$)